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AI for Inclusive Energy Workplaces in Industry and Academia

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Abstract

As the energy sector evolves, creating a truly inclusive workplace is critical. This study explores how artificial intelligence (AI) can be leveraged to embed diversity and inclusion into organizational culture across energy companies and academic institutions. By examining tools such as bias detection algorithms, inclusive language models, and AI-powered mentorship programs, we demonstrate how AI can identify unconscious bias, enhance recruitment fairness, support inclusive policy development, and promote equity throughout organizational structures. This study employs a multi-pronged approach, starting with a comprehensive literature review of classical studies on AI, its growing role in the energy sector, and how it plays a role in a diverse and inclusive workplace in the energy industry. We also analyze current ethical challenges related to deploying AI in energy workplaces and academic environments. Furthermore, we integrate case studies from existing industry practices to compare theoretical frameworks with real-world applications. In addition to this review and analysis, a survey questionnaire was designed and deployed which targeted faculty members in engineering fields on a global scale. The survey gathered responses on how AI tools are currently used in academia, their role in promoting diversity and inclusion, and attitudes toward future opportunities. Preliminary findings revealed that although most participants reported using AI for educational or administrative purposes, a minimal percentage employed these tools specifically for diversity and inclusion, highlighting a notable gap in implementation. This comparative analysis enables us to evaluate how well AI is currently being used to promote inclusiveness and identify gaps that require attention for future implementation and policy design. Preliminary investigations reveal significant ethical dilemmas and a lack of effective inclusiveness when AI tools are applied in recruitment, performance evaluation, and professional development in both the energy industry and academia. The current application of AI often lacks fair representation in training data, leading to biased outcomes. This compromises efforts toward equity and diversity and limits the potential of AI in efficiently unlocking the benefits of more inclusive workplaces. Our findings suggest that a more structured and ethically guided methodology is required to improve the inclusiveness of AI systems and ensure fair outcomes across all levels of organizational decision-making. This study offers a pioneering blueprint for using AI as a tool for

promoting inclusiveness in the energy sector and academia. Unlike existing works, it evaluates the tangible impact of AI tools on equity outcomes, integrates original survey-based evidence from global academic institutions, and provides actionable insights for improving workplace inclusivity and supporting future generations of diverse professionals and leaders.

Keywords: Artificial Intelligence (AI), Diversity and Inclusion (D&I), Workplace Equity, Industry, Academia

Introduction

The energy industry has relied on disruptive technologies to meet the growing global energy demand. From rotary drilling in 1901 to hydraulic fracturing and digital twin modeling in more recent history. However, our industry is also known for its cautious approach to adoption of technology, often constrained by legacy systems, risk-averse cultures, and hierarchical decision-making (Gleckler 2025). For example, intelligent well designs and coil-tubing drilling—now considered standard in many operations—took over a decade to gain commercial traction due to high development costs and market uncertainty (Radelet 2022). The adoption of artificial intelligence (AI) followed a similar trajectory of initial hesitation followed by rapid acceleration. AI applications in the energy industry were largely experimental until the mid-2010s, where it was first adopted in seismic interpretation and equipment monitoring. A dramatic shift was seen in more recent years where AI-related publications in the oil and gas industry increased by 15% per year between 2015 and 2025 reflecting a rise in academic and industrial investment and interest (Póvoas et al., 2025). Today, AI is transforming upstream and downstream operations from real-time reservoir modeling and predictive maintenance to emissions tracking and supply chain optimization (Adsit et al. 2024).

Beyond technical applications, AI emerges as a key tool to foster inclusive workplaces. Inclusive workplaces are an environment where all individuals feel valued, respected, and have equal opportunities to succeed. The energy sector has historically been a male-dominated field. Academia, too, faces issues with bias in areas such as admissions, faculty hiring, and student support programs. Women and minorities remain underrepresented in many STEM fields (Voutyrakou and Skordoulis 2025). As of 2023, women constitute only ~20% of the global technical energy workforce and an even lower share of leadership roles with only 5% of executive roles being held by women (International Energy Agency 2023; Watson 2024). Likewise, women hold only ~28 to 31% of STEM academic positions. In the United States, women earned only ~16.5% of bachelor's degrees in mechanical engineering and ~37.7% in chemical engineering in 2020 (ASCE 2023). Even fewer women reach faculty ranks with ~15 to 20% of tenure-track faculty in these fields being female. Underrepresented racial and ethnic groups are also scarce, with only ~5% African American and 7% Latino representation in engineering professions, and even lower representation in leadership positions at less than 3% (Fry, Kennedy, and Funk 2021).

AI can personalize and adapt instruction to meet diverse student needs. Especially in the field of mechanical engineering, the table below reveals a range of findings from Alghazo, Ahmed, and Bahroun 2025, which relates literature and the integration of AI usage in Mechanical Engineering.

The referenced studies demonstrate how AI may support student-centered learning, encourage personalized learning, and ease the learning process.

Inclusive workplace environments are widely recognized as essential in both industry and academia, where diverse perspectives drive innovation, business value and equitable outcomes (Tawfik and Cuesta 2025; Hamp, Ryan, and Carreras 2020). Companies are using AI and machine learning to identify bias in hiring practices, automate unbiased job screening, analyze employee sentiment, personalize mentoring, and improve representation across teams (Adsit et al. 2024). This marks a significant cultural shift, as energy companies begin to recognize that digital transformation must include not only operational efficiency but also organizational equity. This necessitates the responsible use of AI. Researchers emphasize that generative AI often produces predictable, 'basic' outputs shaped by its training data, raising concerns about

diversity and originality in both industry and higher education, therefore without intervention "that's what an AI is likely to return." (Sturgill 2025). This paper provides a review on how AI can be responsibly used to promote inclusivity in both industry and academic institutions. We examine the current state of AI applications in this domain, best practices and strategies, case studies of successful implementations, challenges, and future opportunities.

Methodology

This study adopts a qualitative literature review methodology to investigate how artificial intelligence (AI) is being utilized to support diversity and inclusion (DI) in the energy workplace both in industry and academia. The approach was selected to identify existing trends, practices, technologies, challenges, and future opportunities.

Literature Selection Criteria

We have used a two-pronged approach. In the first approach, we conducted a qualitative literature review to explore how artificial intelligence (AI) supports diversity and inclusion (DI) in the energy workplace, both in industry and academia. The objective was to identify existing trends, practices, technologies, challenges, and future opportunities. Relevant literature was sourced from IEEE Xplore, ScienceDirect, Google Scholar, OnePetro, Frontiers, and ERIC, with a focus on AI in petroleum, chemical, and mechanical engineering, as well as diversity and inclusion. Publications from 2018 to 2024 were prioritized.

Survey Design and Distribution

In the second approach, we conducted an online survey to understand the current state of AI and DI in academia worldwide. After an extensive review of existing literature on AI in academia, survey questions were developed ensuring alignment with the established frameworks. The survey was created around four main themes: (1) current usage of AI in teaching and administration; (2) the extent of AI adoption for diversity and inclusive purposes; (3) perceived benefits and obstacles; and (4) anticipated applications in the future. Pilot testing was conducted amongst a small group of faculty members to refine the questions clarity and to reduce ambiguity before sending it worldwide.

Universities were selected using a combination of targeted outreach and convenience sampling, including the top universities worldwide based on well-known rankings such as QS, THE, and ARWU rankings. Priority was given to mechanical and chemical engineering programs. Using the university websites faculty information and mailing lists, the survey was sent to the engineering department heads and chairs as the first point of contact, which then turned to the associate professors. This is because they are most directly involved in the teaching and know their departments AI and diversity initiatives for their program. Faculty from Asia, North America, South America, Europe, Africa, Oceania, and the Middle East were contacted to ensure geographic diversity. The survey was distributed electronically using Google Forms and shared through institutional email lists. The respondents were informed that it was voluntary and that participation would remain anonymous, reducing the chances of self-selection bias related to professional standing or reputation.

Future Directions

Future iterations of this questionnaire will include expanding the sample size across more universities as well as expanding into more engineering disciplines such as Electrical engineering, Industrial engineering and more. In addition, cross-sector comparisons will be explored by including participants from the energy industry to see the comparison between industrial and academic usage of AI for diversity and inclusion.

Results and Discussions

This section presents the findings from both the literature and the preliminary survey results, focusing on how AI is currently being applied to support diversity and inclusion in academia and industry. The analysis is split into two parts: first, the applications of AI in academia, particularly in engineering, where mentoring platforms, bias detection and adaptive learning is transforming student support and teaching; and second, the applications of AI in the energy industry, where digital transformation initiatives intersect with workforce inclusion and equity. The survey received 26 responses from faculty across 15 countries. Preliminary results revealed that 70.8% of respondents have used AI tools in some teaching capacity, however only 8.3% have applied them for diversity or inclusive purposes. This gap between AI usage and D&I specific applications reflects the trends noted in this paper. The results will provide insights into current adoption patterns and help highlight gaps between general AI use and its targeted application for equity goals.

Results: Academia

AI Tools for Inclusive Teaching and Learning. The cited papers in [Table 1](#) show that AI can facilitate learning, promote personalized learning, and advocate for student centered learning. For instance, intelligent tutoring systems monitor each learner's performance and provide tailored feedback and tasks ([Alghazo, Ahmed, and Bahroun 2025](#)). Such personalized AI tutors boost confidence and independence, offering real-time guidance in areas of strength or struggle ([Stand Together 2025](#)). This approach enhances the educational experience by fostering greater creativity and efficiency, while simultaneously increasing student motivation, engagement, and preparedness for their future careers.

Table 1—AI usage in Mechanical Engineering (adapted from [Alghazo, Ahmed, and Bahroun 2025](#))

Theme	Author	Focus
Artificial Intelligence in MEE via assessment context	Kahangamage and Leung (2019)	Redesigning engineering design courses to improve student learning outcomes and teaching strategies.
	Shyr et al. (2019)	To gauge the educational impact of incorporating AI into robotics design projects, a framework of assessment indicators is needed to evaluate its specific influence on engineering students' learning outcomes.
	Kuzilek et al. (2021)	Using an AI algorithm to forecast student performance by analyzing their behavior during exams.
Assessment tools in CAD	Pando Cerra et al. (2023)	Examining how integrating the self-assessment tool TrainCAD into CAD teaching methods benefits learners and influences their academic achievement.
	Jaakma and Kiviluoma (2019)	Introducing two new automated online assessment tools to aid in developing both procedural and strategic CAD skills.
AI in classification and categorizing in MEE	Rodriguez-Martin et al. (2019)	Discusses integrating short CFD simulations into fluid mechanics education and highlights the application of machine learning in teaching a multidisciplinary student population.
	Belapurkar et al. (2019)	Implementing an automated system for assigning project themes to improve student learning, specifically within the context of robotics competitions.
AI aids in Assisting students & Facilitate learning	Huang et al. (2023)	Investigates the integration of artificial intelligence and design, examining its historical evolution, current applications, and the challenges designers face in adopting AI tools.
	Auerbach et al. (2019)	Investigating how robotics, specifically RoboGen, supports inquiry-based learning by enriching student exploration and experience in AI and robotics education.
	Cai et al. (2021)	Introducing a chatbot that explains mathematical concepts to facilitate personalized learning.
	Chen D. et al. (2023)	Integrating precision machinery, artificial intelligence, and customized learning materials into the curriculum for precision measurement courses.
	Clark and Clark (2018)	Creation of an adaptive learning tool specifically engineered for thermodynamics instruction.

Theme	Author	Focus
	Bi (2020)	Using Natural Language Processing (LNP) to translate core STEM concepts into clear English text.
	Hsieh and Li (2018)	Developing a system to help students learn G-code programming and understand the essential hardware concepts required for CNC operations.
	Liu et al. (2021b)	Applying AI techniques such as BP neural networks and hill climbing algorithms to improve mechanical engineering instruction and training in higher education.
	Lin et al. (2020a)	Investigating how AI can be integrated into machinery manufacturing education to enhance teaching and learning.
	Tsai et al. (2018)	Creating an intelligent mechanical lab that uses Ai to facilitate advanced educational experiences.
	Lin et al. (2020)	

AI can also help instructors create more inclusive materials. For instance, ChatGPT and similar models have been shown to flag biased or stereotyped content in classroom activities and suggest neutral alternatives (Voutyrakou and Skordoulis 2025). Survey data supports this potential as seen illustrated in the figure below with approximately 68.4% of respondents reported using AI tools for academic support (e.g., learning platforms, curriculum design, tutoring, grading, etc.).

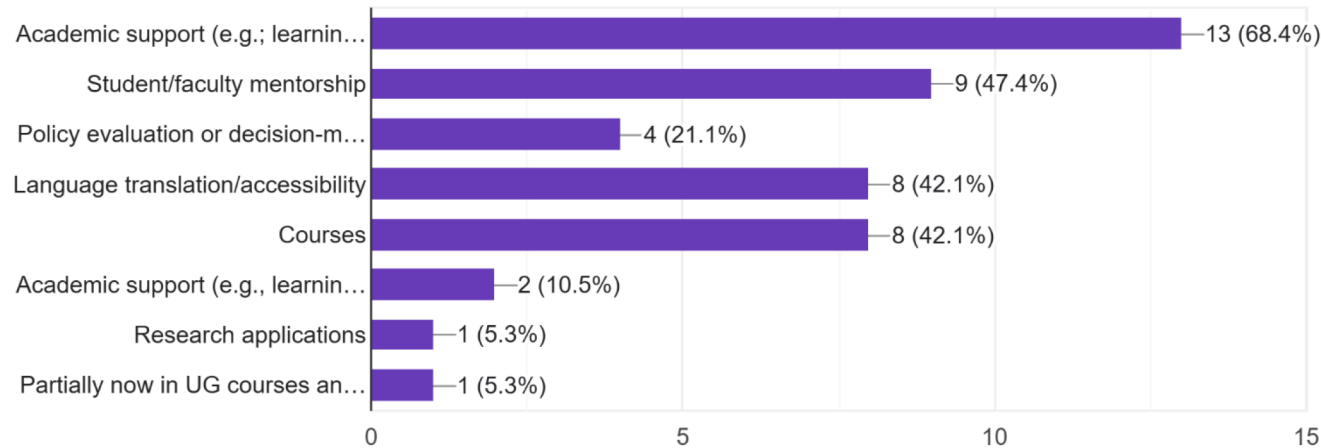


Figure 1—Survey Question: "What is the primary purpose of AI use in your department?"

Furthermore, instructors can use AI assistants to simplify language, explain idioms, or rephrase instructions, helping English learners and neurodivergent students understand more easily (Sturgill 2025). Emerging tools can even generate alt-text for images and transcripts of lectures, making content accessible to students with visual or hearing impairments (Gonzalez et al. 2024).

Adaptive Tutoring and Analytics. AI-driven tutoring systems can personalize instruction by tracking each student's performance and delivering tailored practice or feedback. For example, platforms like Khan Academy and ALEKS use machine learning to diagnose skill gaps and adapt the next problems to the student (Alghazo, Ahmed, and Bahroun 2025). In mechanical engineering, recent reviews confirm that AI applications include personalized learning experiences and smart tutoring systems, which can boost student motivation (Alghazo, Ahmed, and Bahroun 2025). AI-based analytics can also flag when a student is struggling (regardless of background), allowing early intervention by instructors. In general, adaptive learning has been shown to increase students' confidence and independence (Alghazo, Ahmed, and Bahroun 2025). Such systems could be used in ME/ChE courses to help underprepared or nontraditional students stay on track.

Bias Detection and Inclusive Content Review. AI can analyze course content to identify biased or exclusive language. For instance, a recent study used ChatGPT to review educational materials for gendered or stereotypical terms and then rewrote them neutrally (Voutyrakou and Skordoulis 2025). Voutyrakou & Skordoulis (2025) found that ChatGPT could highlight unconscious gender bias in engineering curricula and help generate more inclusive problem statements. Similarly, tools are emerging that scan syllabi or exams for problematic cultural references or language that might discourage certain groups. By flagging and suggesting edits, these AI tools help instructors produce more neutral, inclusive content.

Inclusive Language Models for Clarity. Beyond bias detection, generative AI can simplify and rephrase material to accommodate diverse backgrounds and reading levels. Educators are encouraged to use GPT-style assistants to "generate alternative explanations for difficult concepts using examples or analogies that students are familiar with," tailoring prompts to specific audiences (Gonzalez et al. 2024). For example, an instructor might ask an AI to reword a technical problem in plain language for an English-language learner, or to create family-oriented analogies for abstract topics. This flexibility can support students who are first-generation college attendees, non-native speakers, or who have varying academic preparation. Early evidence suggests students using AI paraphrasing tools improve comprehension and demystify "hidden rules" of academic language (Gonzalez et al. 2024).

Accessibility and Assistive Tools. AI greatly aids accessibility: automated captioning and image description make STEM content available to students with sensory impairments. Current AI tools can "generate alt text or long descriptions for images" and provide transcripts of lectures (Gonzalez et al. 2024). In practice, an engineering instructor could use AI to auto-generate captions for a video of a laboratory demo or to describe complex diagrams. AI can also summarize lectures or slides into outlines for review (Gonzalez et al. 2024). By lowering these access barriers, AI helps students with disabilities or diverse learning preferences engage fully in ME/ChE courses.

AI-Enhanced Mentoring and Community Support. Beyond classrooms, AI can expand mentoring and peer support, especially for underrepresented students. Projects like CareerFair.ai use conversational AI to simulate realistic chats with STEM professionals from diverse backgrounds. By embedding videos of actual minority and first-generation STEM mentors into virtual agents, underrepresented students can ask career questions any time. As one researcher explains, this gives students "each mentor's advice in a personalized way to many more students" than in-person mentoring allows (Okado, Nye, and Swartout 2022). Likewise, AI chatbots can provide anonymous venues to discuss identity concerns (e.g., disability self-advocacy) without fear of judgment. In nursing education, AI-based role-play simulates conversations with patients of diverse backgrounds, allowing students to practice cultural competence in a low-pressure setting. The table below summarizes these AI tool categories and their relevance to D&I in engineering education.

Table 2—AI Tools and D&I Applications in Engineering Education.

AI Tool/Category	Diversity Function	Examples/Notes
Adaptive Learning Platforms	Personalized practice/feedback; early alerts for strugglers	E.g., Khan Academy-style systems; AI-driven tutoring in core ME/ChE courses.
Bias Detection (NLP)	Identify exclusionary language or biases in materials	E.g., ChatGPT-based analysis of syllabi for gendered or cultural bias
Language Clarity (GPT)	Simplify/translate content; multiple reading levels	E.g., AI rephrasing technical problems for ESL or first-gen students
Accessibility Generators	Alt-text for images; transcripts/summaries	E.g., AI captioning of lecture videos, diagram descriptions in ME/ChE classes

Similarly, AI-powered peer networks scale access to learning communities. Platforms like Schoolhouse.world integrate AI-driven diagnostics and tutor matching, connecting learners across 100+ countries and planning to reach millions ([Stand Together 2025](#)). Such systems democratize access to high-quality tutoring and build supportive global communities.

Trends and Challenges

Recent years show rapid uptake of AI in education broadly. A 2024 survey found 84% of U.S. higher-education professionals use AI tools in work or personal life (up 32 points from 2023), and 93% expect to expand AI use in the next 2 years ([Gonzalez et al. 2024](#)).

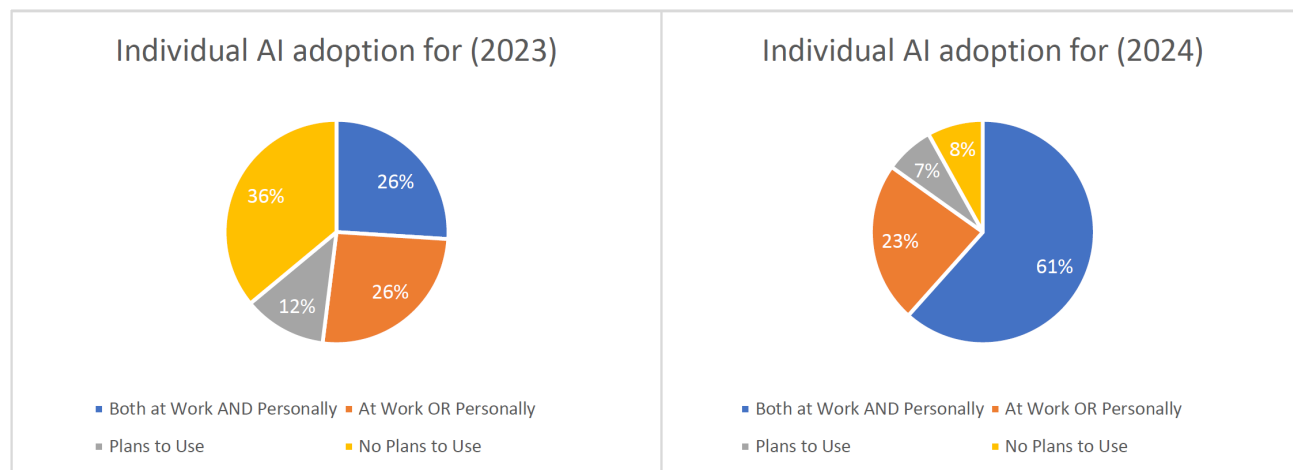


Figure 2—Trends in AI adoption. (adapted from [Ellucian 2024](#)), "2023 & 2024 Survey Question: Which of the following best describes how you have used artificial intelligence (AI)?"

Many administrators note AI's promise for efficiency and personalized support ([Gonzalez et al. 2024](#)). In engineering education specifically, systematic reviews identify emerging themes: besides personalization, generative AI is being explored (e.g., ChatGPT in design problems) ([Alghazo, Ahmed, and Bahroun 2025](#)), as are analytics for assessment and simulation. For D&I, there is growing recognition that AI can democratize access. for example, by delivering high-quality tutoring to any student or by synthesizing inclusive content at scale ([Voutyrakou and Skordoulis 2025](#)).

While AI offers powerful new tools, it also poses risks if left unchecked. Data-driven models can inherit biases from historical patterns. A majority of educators worry about AI's ethical implications: in one survey, 49% of respondents cited bias in AI models as a top concern (up from 36% a year earlier), and 59% cited data security/privacy ([Gonzalez et al. 2024](#)). Notably, 78% worried AI could harm academic integrity ([Gonzalez et al. 2024](#)). These worries translate into barriers: faculty may resist new tools if they mistrust their fairness or lack training. Critically, algorithmic bias is a central issue. A recent University of Washington study found that state-of-the-art language models ranked resumes with white male names dramatically higher than those with Black or female-associated names ([Wilson and Caliskan 2024](#)). This "significant racial, gender and intersectional bias" in AI hiring tools underscores the danger: an AI system trained on unrepresentative data will exclude marginalized candidates.

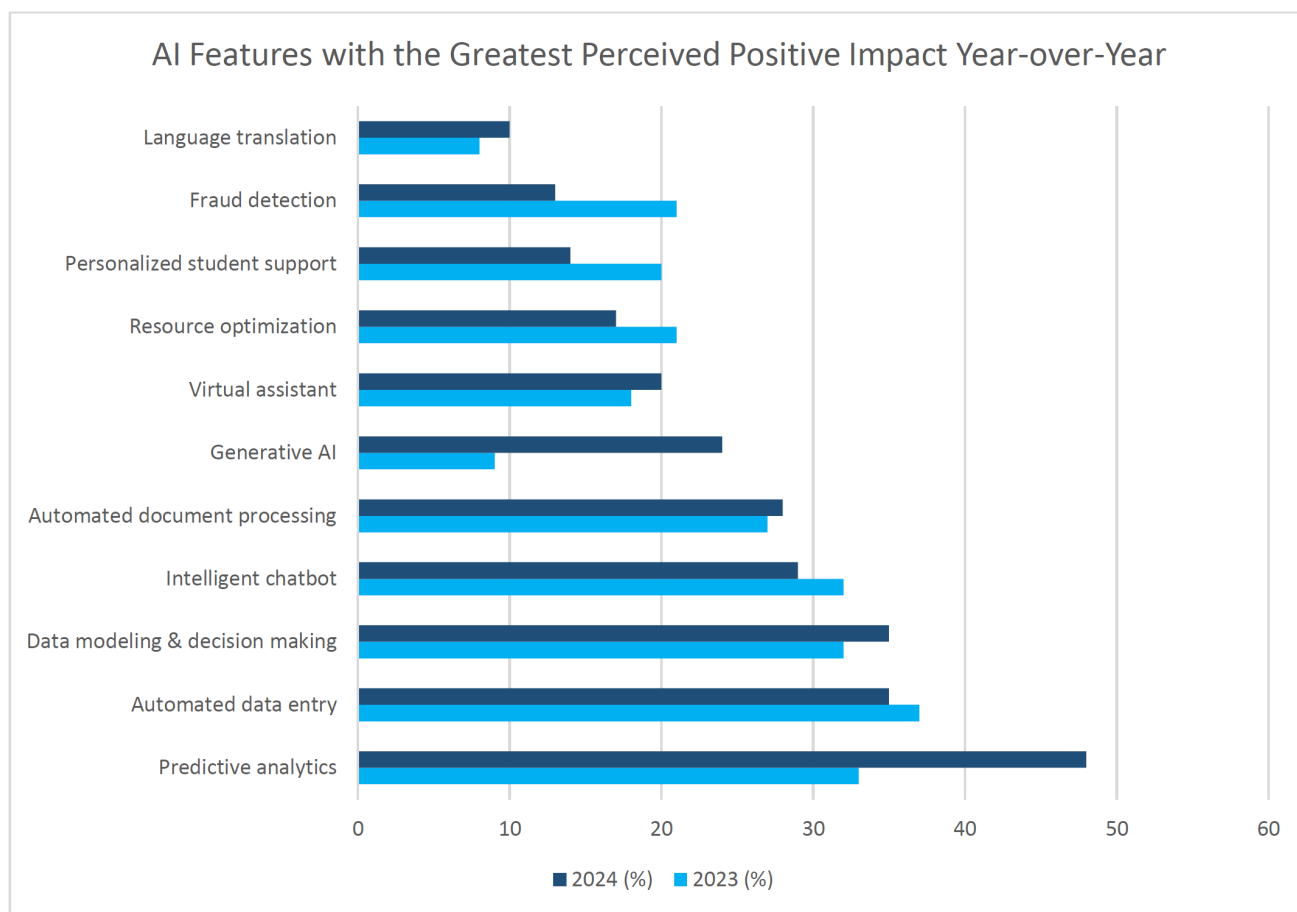


Figure 3—Perceived Positive Impact of AI. (adapted from Ellucian 2024), "2023 & 2024 Survey Question: What AI functionality would have the greatest positive impact on your department?"

Other barriers include resource constraints. Deploying advanced AI tools requires infrastructure (cloud access, computing power) and faculty time for training. Under-resourced institutions or small departments may find it hard to adopt sophisticated systems. Additionally, there is a risk of digital divides: students from disadvantaged backgrounds might have less access to the latest technology, undermining equity if not managed carefully. Finally, policy and guidelines are still catching up; without clear institutional support and ethical frameworks, departments may be reluctant to use AI even if it could advance D&I.

In summary, AI can bolster inclusion in STEM education and campus culture by automating equitable practices, but only if institutions build in checks and human judgment. By combining adaptive AI tutors, unbiased language models, and proactive mentorship programs, universities can move toward truly inclusive environments. As our literature review will show, leveraging these technologies responsibly offers a roadmap for AI-driven equity across academic institutions.

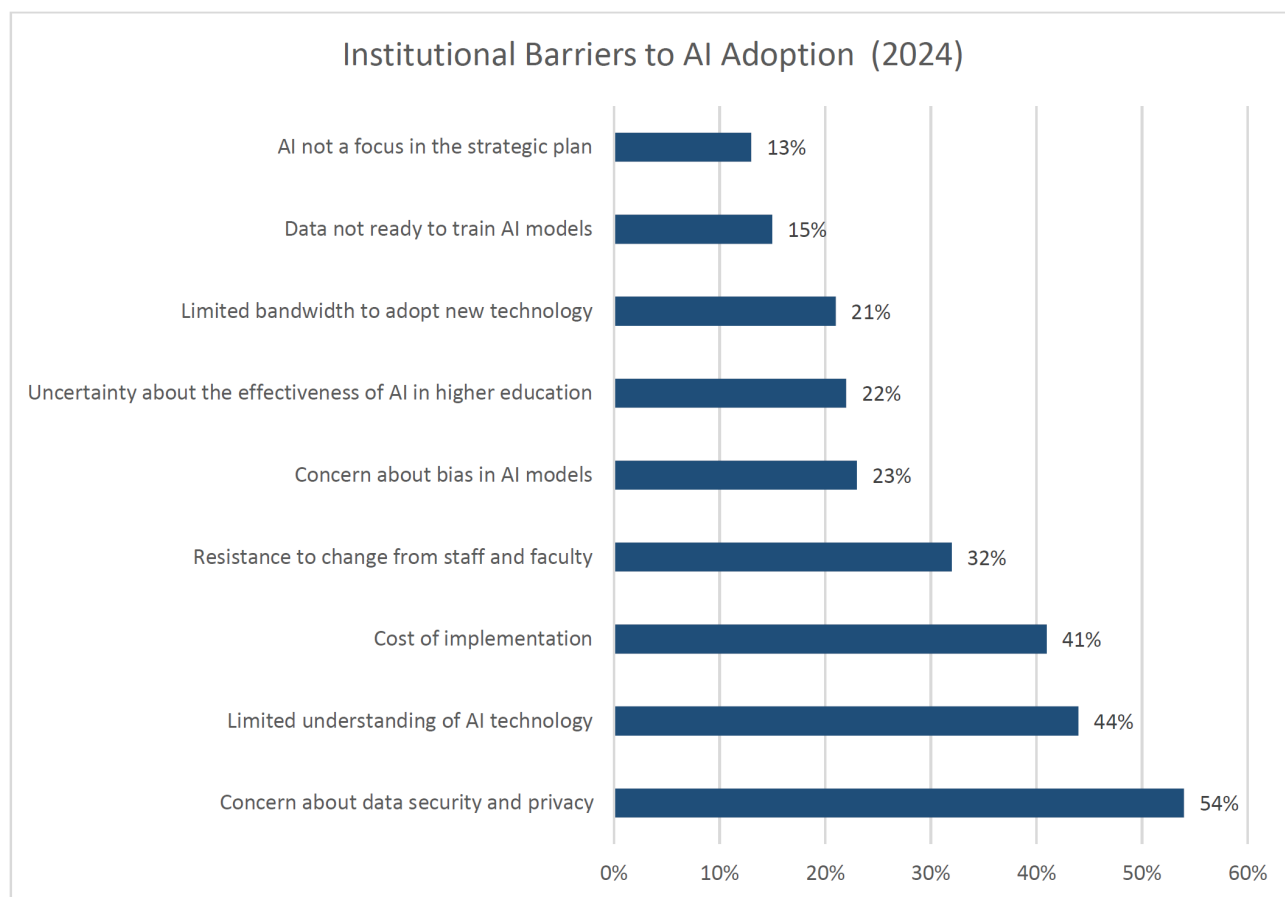


Figure 4—Barriers to AI adoption. (adapted from [Ellucian 2024](#)), "2024 Survey Question: What are your institution's main barriers to adopting AI? Select up to three barriers."

Case Studies and Institutional Practices

There are few published case studies specific to mechanical or chemical engineering that directly combine AI with D&I. However, some broader initiatives are instructive:

Chemical Engineering (AICHE/IDEAL Workshop). The American Institute of Chemical Engineers (AIChE) convened a national workshop on diversity equity and inclusion (DEI) for ChE academic leaders (AIChE 2022). While not AI-focused, this IDEAL (Inclusion, Diversity, Equity, Anti-racism, Learning) workshop signals departmental-level attention to D&I (women –38% of ChE undergraduates) (AIChE 2022). AI tools could complement such efforts by, e.g., auditing curricula for bias or personalizing support in ChE classes.

Mechanical Engineering (SDSU DEI Plan). San Diego State University's ME department set explicit hiring goals (increasing female faculty from 13% to 18%) and outlines inclusive hiring strategies (San Diego State University 2023). Again, while not AI-specific, it shows departmental commitment. Such programs may integrate AI in future (e.g., using AI screening tools judged for fairness).

AI Mentoring Platform (CareerFair.ai). Though covering all STEM, the CareerFair.ai project at Cal State Fullerton/USC is a notable AI-for-D&I application (Okado, Nye, and Swartout 2022). It uses conversational AI to give diverse students virtual access to STEM mentors from similar backgrounds. Early evaluations show high student satisfaction, suggesting a path to scale mentoring for engineering students nationwide.

Discussion Groups and Tool Development. At many universities, committees on teaching or computing are exploring AI use-cases. For example, some institutions develop custom GPT assistants trained on course

syllabi (Gonzalez et al. 2024). Formal published reports are scarce, but campus workshops often address AI literacy, implicit bias training, and inclusive syllabus, the groundwork for later AI adoption.

In summary, while explicit AI + DI case studies in mech/chem are limited, general higher-ed practices (e.g., AI for accessible content (Gonzalez et al. 2024) or AI mentoring (Okado, Nye, and Swartout 2022)) can be adopted by engineering departments. Future work may report pilot implementations (e.g., an ME faculty using ChatGPT to rephrase problem sets for clarity).

Results: Industry

AI adoption in the energy industry has accelerated in recent years, primarily for operational efficiency and safety. Increasingly, these AI and digital transformation initiatives are intersecting with workforce development and DI goals. For example, the use of AI is reshaping the energy workforce skillset (Randolph 2025). This workforce shift may be viewed as a threat eliminating more traditional roles, but also as an opportunity where companies use AI to actively reskill and upskill their employees through AI platforms, ensuring long-time employees are not left behind in the digital transition. Many workers themselves are eager to adapt - a 2024 global survey showed 62% of employees expect AI's impact on their job to be positive, and 57% want their employer to provide AI training to help them grow. This sentiment reinforces that inclusive upskilling is not just top-down but also demanded bottom-up (Machuel, 2024).

A successful example is Shell investing in democratizing AI skills among their employees, training staff in AI to empower a broader base of workers as "Citizen Data Scientists," effectively turning engineers and scientists into citizen developers who create their own AI-driven solutions (Shell 2021). Through hackathons and a global learning community (Shell.ai), Shell enabled hundreds of these citizen data scientists to build bespoke AI applications (for instance, optimizing chemical plant operations), thereby broadening digital expertise beyond just the data science team. Similarly, Chevron has partnered with Accenture and Microsoft and invested heavily in creating a global workforce that is capable of unlocking the full potential of responsible AI adoption, through their "Digital Scholar Program" and "Citizen Development Program." (Abinader, Guthrie, and Herrington 2025; Kitishian 2025). Its "Digital Scholar Program" sponsors selected employees worldwide (from the US to Nigeria and Australia) to take a one-year leave and earn a master's degree in data science or systems engineering at MIT or Rice University. In parallel, Chevron's "Citizen Developer" initiative trained over 2,800 employees in low-code app development, supported by a community of hundreds of mentors, to enable any worker (regardless of formal IT background) to create technology solutions. Such initiatives indicate a commitment to an inclusive digital transformation, where technology is used to include and elevate existing employees across ages, roles, and geographies. Even national oil companies are embracing this trend: for example, Saudi Aramco's digital arm launched a major program in 2024 to build an "AI-ready workforce" in line with the country's Vision 2030. Aramco's partnership with Accenture leverages a generative AI-powered learning platform to upskill the company's entire workforce in AI, using personalized content in both Arabic and English (Halabi & Alrajab, 2024). These global efforts underscore that energy companies in the Middle East, Europe, and the Americas alike see empowering their people with AI skills as a strategic imperative, not just a CSR effort. In fact, a U.K. industry survey highlighted that lack of digital talent is a key risk for energy firms' digital plans, strongly urging more employee training and skill development initiatives.

AI is also being utilized in human resources and talent management to promote workforce inclusiveness in energy companies. AI tools are being piloted for unbiased recruitment, tailored talent development and mentoring programs, and monitoring workplace culture and sentiments. A 2024 review of AI in oil and gas recruitment found that AI technologies are transforming hiring processes through automating routine tasks, improving candidate matching, and enabling data-driven decisions, while reducing unconscious biases in screening and selections (Odili et al. 2024).

AI-Driven Bias Detection & Mitigation in Recruitment. Besides the obvious benefit of improved efficiency, using AI-driven recruitment tools can make hiring fairer. One of the most immediate applications is in reducing bias during recruitment and candidate selection. AI- powered recruiting tools can analyze resumes and applications blind to demographic identifiers, focusing only on skills and qualifications. For example, algorithms can strip out names, gender, age, and other identifying information from resumes during first-round screening. This "blind hiring" approach, augmented by AI, has led to demonstrably more diverse interview pools in pilots, so that decisions are based on merit rather than unconscious biases, ensuring candidates from underrepresented groups get a fair chance. By training algorithms on large datasets of successful employees' skills rather than on who was hired in the past, which could perpetuate bias, companies like BP and Siemens Energy have reported improvements in identifying non-traditional candidates who nonetheless fit the job requirements. AI can also be applied to internal promotions and performance evaluations - analyzing performance data without favoritism - to ensure equitable career advancement opportunities within the organization. Beyond screening existing applicants, AI tools can be used to identify where job postings might be unintentionally biased, for example, by flagging masculine-coded language that deters female applicants and instead suggest more inclusive wording ([Talent Management Institute 2023](#)). AI-driven outreach tools can also search for qualified individuals in diverse professional networks or academic circles, helping recruiters in both industry and academia reach broader talent pools. That ensures that the talent pipeline itself becomes more diverse before individuals even apply.

Enhancing Candidate Experience and Retention. AI chatbots and virtual assistants can provide timely, personalized communication with job candidates or prospective students, ensuring everyone gets equal information and quick responses to their queries. These tools can also help with sending follow-ups, scheduling interviews (preventing delays that might lose a candidate to a competitor), and gathering candidate feedback. By analyzing this data, companies can spot if certain demographics are having worse experiences (e.g., longer wait times for replies, or technical issues with online assessments) and then fix those process gaps. This consistent, unbiased engagement leads to a positive candidate experience for all, not just those who might feel empowered to email a recruiter directly. Candidates who feel respected and informed during the recruitment process are 38% more likely to join and stay ([Talent Management Institute 2023](#)).

AI for Training and Personalized Learning. Continuous development is key to inclusion, as it ensures all employees or students have access to growth opportunities that allow them to succeed. AI has revolutionized training through personalization. In corporate environments, AI-driven learning platforms create custom training programs tailored to each employee's needs, pace, style and aiming point ([Talent Management Institute 2023](#)). This is beneficial in the energy industry where the workforce ranges from field technicians without formal degrees to highly educated engineers. AI can tailor content accordingly, so everyone can upskill effectively for the benefit of the company as well as the employee. For example, if a field worker needs to learn about a new AI-powered drilling software, an AI tutor might adjust the training to that worker's baseline knowledge, offering more foundational lessons at first and practical "how to" content. Conversely, an engineer or subject matter expert who is more familiar with AI-driven drilling software might get an accelerated module. Personalized learning promotes inclusion by meeting individuals where they are, closing skill gaps that often align with demographic disparities.

Improving Accessibility for People with Disabilities. The World Health Organization (WHO) estimates that over 1 billion people globally have some form of disability ([Talent Management Institute 2023](#)). AI-driven accessibility solutions have a profound impact on removing barriers and allowing an otherwise untapped talent pool to be included. For example, AI-powered text-to-speech and speech-to-text tools (e.g.: live transcription during virtual meetings) enables visually impaired or hearing- impaired colleagues to participate fully during meetings and conversations. Computer vision AI also provides visual description

of images or physical environments to vision-impaired colleagues. By embracing AI for accessibility, organizations demonstrate that everyone is valued, fostering a culture where differences are accommodated rather than seen as an obstacle.

Discussions

Despite rising interest in AI adoption in literature review, survey results suggest that intentional use of AI for diversity and inclusion remains limited or unclear in many academic contexts. As shown in [Figure 5](#), only 8.3% of respondents confirmed that AI tools in their department were used with a specific D&I focus. In contrast, 41.7% reported no such use, and a majority (50%) were unsure. This uncertainty points to a possible gap between general AI usage and purposeful, equity-oriented applications. Nevertheless, some survey participants expressed strong support for AI's potential for inclusivity. One responded stated, "I think AI is very important to support diversity and inclusion in higher education. So, it enhances teaching and learning and improves student outcomes." This perspective reinforces the idea that while current applications of AI may be limited, there is a clear recognition of AI's potential to make education more equitable. This highlights a need for awareness, training, and clearer strategies from institutions to ensure that AI tools are actively supporting inclusive goals.

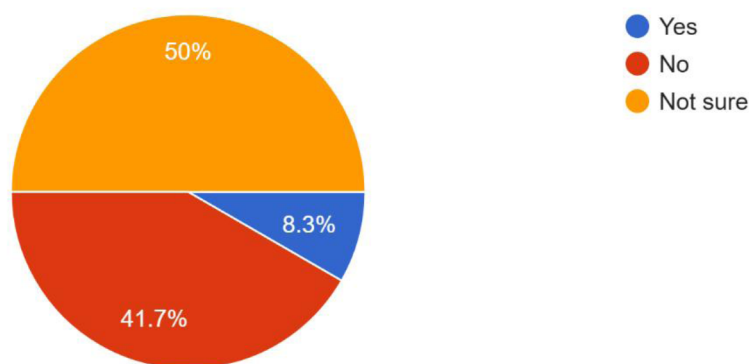


Figure 5—Survey Question: "Are any of the AI tools used with a specific focus on promoting diversity and inclusion?"

Furthermore, according to survey results, the integration of AI ethics and bias awareness into engineering curricula appears uneven. According to [Figure 6](#), no respondents reported that courses that explore AI in the context of ethics, bias or social impact are required in their programs. Only 12.5% indicated that such content is available as an elective or within another course. Another 29.2% indicated that these topics are being considered, while 16.7% reported no offerings of such AI courses at all. Surprisingly, 41.7% of participants were unsure, suggesting either lack of institutional emphasis on this area or limited transparency. These findings support calls for departments to adopt ethical AI courses into core engineering education, especially of future engineers are to build bias-aware, inclusive systems.

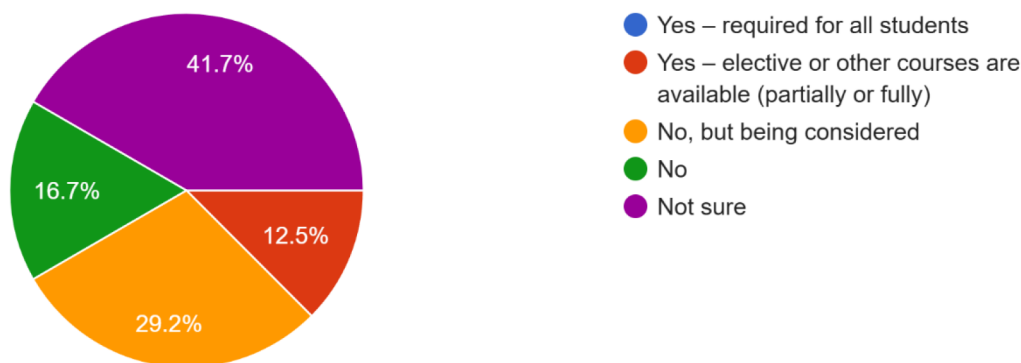


Figure 6—Survey Question: "Are there courses offered at your Department/Program that explore AI in the context of ethics, or social impact?"

When asked about the level at which AI is most actively integrated into academic programs, responses were mixed. As illustrated in Figure 7, 45.8% of participants noted that AI is implemented equally across all program levels. This suggests a broad-based approach to adoption. At the same time, 20.8% established AI activity at the bachelor's level and 12.5% at the master's level. Another 16.7% stated that the question was not applicable to their institution. These findings imply that some institutions take a system-wide approach, while others may still focus AI adoption on specific departments or programs. This variance in program adoption presents an opportunity to standardize inclusion-focused AI practices across all levels of engineering education.

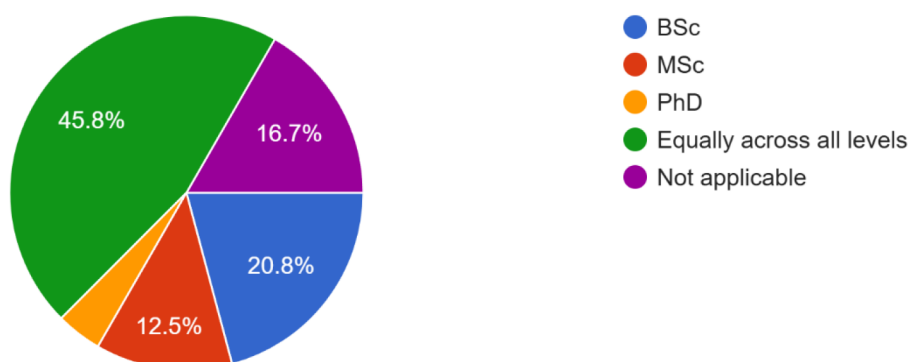


Figure 7—Survey Question: "At what program level is AI most actively integrated?"

AI has significant potential to support diversity and inclusion in mechanical and chemical engineering education. The literature shows an expanding toolkit: from adaptive tutoring that can help historically underperforming students, to NLP tools that sanitize course material, to AI-driven networking that broadens mentorship. These tools can reduce some inequities. for example, by giving English-language learners simplified explanations or providing continuous feedback to students who might otherwise "fall through the cracks."

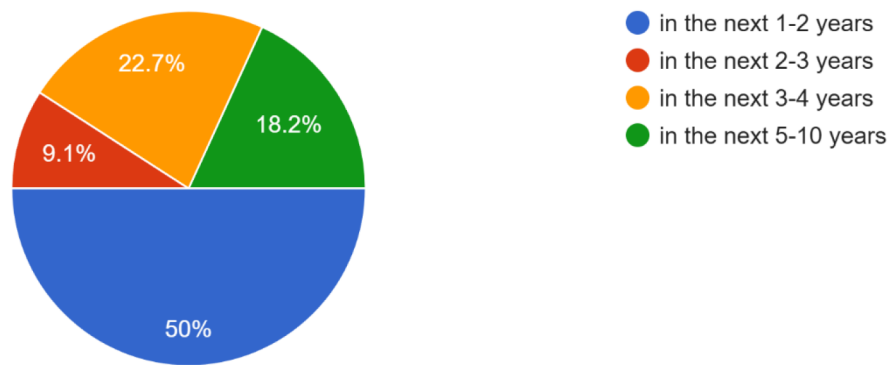


Figure 8—Survey Question: "When do you plan to expand AI use for D&I?"

A majority of survey respondents (50%) expressed plans to expand their use of AI within the next one to two years, as displayed in Figure 5. This growing interest reflects a readiness within the academic engineering community to engage more deeply with AI tools. It also presents a timely opportunity for institutions to guide this momentum toward inclusive applications such as bias-aware content generation, accessible learning platforms, and adaptive support tailored for diverse student populations. Rather than focusing on concerns or limitations to AI use, survey respondents offered constructive views on AI's potential impact.

According to Figure 9, 70.8% of respondents highlighted 'enhanced learning and teaching' as the biggest motivation for adopting AI. 25% cited "supporting diverse and inclusive learning environments" such as captioning, alt text generation, or simplified content delivery. Another 25% selected "efficiency and cost reduction" as a key motivation. In principle, as several reviewers note, "by carefully applying AI technologies, educators can find ways to embrace diversity, build community, and foster belonging" (Gonzalez et al. 2024).

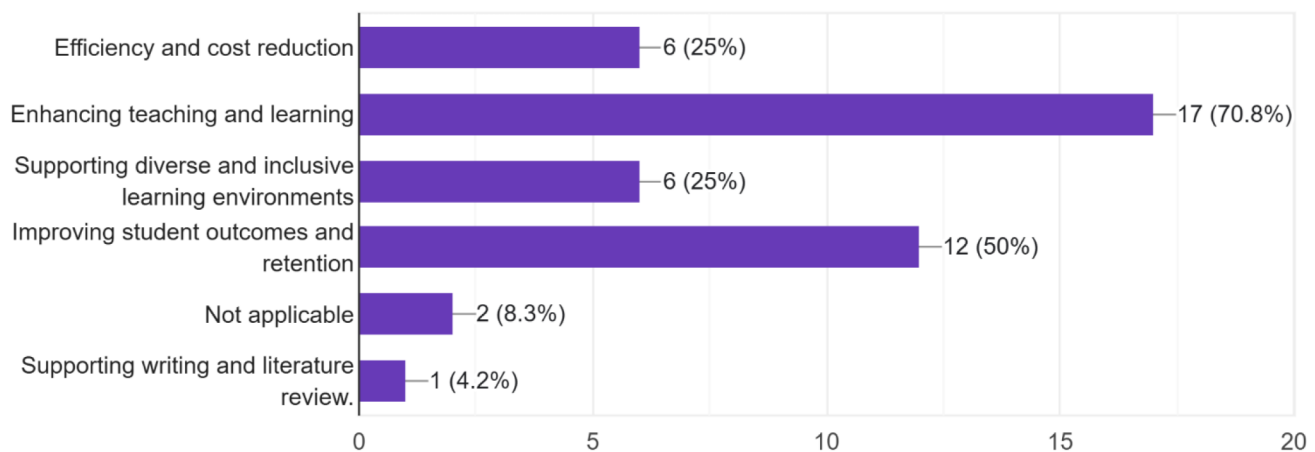


Figure 9—Survey Question: "What is your Department/Program's biggest motivation for adopting AI?"

However, realizing these benefits requires vigilance. The same biases that afflict society can be baked into AI unless addressed. In engineering contexts, it will be critical to evaluate AI tools for bias before wide deployment. This may involve curating training data (e.g., including diverse examples), using fairness-aware algorithms, and involving diverse stakeholders in tool development. Moreover, professional development is needed: instructors and department leaders should be trained both in the pedagogical uses of AI and in its pitfalls (ethical, legal, technical). Guidelines and policies (at institutional and accreditation levels) will shape how AI tools are permitted in classrooms and hiring processes.

This need for ethical oversight is echoed in the open-ended section of the survey, where one respondent emphasized: "To be effective and ethical, these systems must be transparent, regularly audited for bias, and designed with privacy and equity in mind." This perspective reinforces the importance of not only adapting AI tools but doing so with a clear and ethical framework centering inclusiveness from the start. Transparency in AI algorithms and how decisions are made, whether in admissions, course content, or grading, should build trust among faculty and students alike. Regular and frequent audits can ensure that these AI systems remain aligned with institutions' diversity goals over time. Ultimately, ethical implementation of AI systems in academia should be proactive rather than reactive. Equity considerations should be embedded at every stage from system design and data curation to classroom evaluation and integration. This way, institutions can ensure that AI becomes a force for inclusion rather than reproducing existing disparities.

Conclusions

Key Findings

- Low intentional AI use for D&I: Only 8.3% of respondents reported their department uses AI specifically for diversity and inclusion while 41.7% said no and 50% were unsure,
- Lack of required AI ethics and bias courses: No programs require such courses; 12.5% offer them as electives, 29.2% are considering them, and 41.7% of respondents were unsure.
- Broad but uneven AI integration: 45.8% reported AI use equally across all program levels, with the remainder concentrated mainly at BSc (20.8%) and MSc (12.5%) levels.
- Planned growth in AI adoption: 50% plan to expand AI use within the next 1 to 2 years, signaling an opportunity to direct adoption toward inclusive applications.
- Main motivations for AI adoption: 70.8% cited enhanced learning and teaching, 25% supporting inclusive learning environments, and 25% improving efficiency/cost reduction.

In conclusion, the use of AI to embed D&I in engineering academia is an emerging and interdisciplinary field. Key trends point to adaptive and generative AI as the most promising areas, while major challenges remain around bias and trust. Mechanical and chemical engineering programs have much to gain by engaging in these tools but must do so thoughtfully. By synthesizing current research and practice, this review aims to guide engineering educators and researchers toward effective, equitable AI integration in STEM education.

The use of AI in the energy industry suggests AI can be a powerful enabler of inclusion when consciously applied - from hiring a more diverse workforce, to training and elevating existing employees, to empowering those with different abilities. Rather than AI replacing people, the narrative in leading energy companies is increasingly about AI augmenting people - allowing them to do more and focusing on empowering everyone, not just a tech-savvy few. This is a marked shift in mindset: technology projects in energy are no longer only judged on efficiency gains, but also on how they support the company's diversity and inclusion objectives. By integrating AI with inclusive strategies, the energy sector is taking steps toward a future where advanced technology and an inclusive culture reinforce each other.

Future research should include empirical studies in mechanical and chemical engineering settings: for example, piloting an AI teaching assistant in a thermodynamics course or testing an AI syllabus analyzer in a process design class. Surveys and experiments could measure whether AI tools improve retention or sense of belonging for underrepresented students in these majors. Literature review provided here offers a framework: sections on "tools," "barriers," and "cases" can accommodate new data. Once original surveys or experimental data are available (as planned), they can be slotted into this structure, with new figures or tables illustrating outcomes across departments.

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